

Mr. Milk AI × MilkMaster — Integration Brief

A working dairy-intelligence platform, ready to connect to MilkMaster. What we built, how it helps, and what we need.

For: Director (Mr. Milk)

MilkMaster — Director

MilkMaster — Technical Engineer

In one line: Mr. Milk has already built a live AI analytics platform on top of MilkMaster's sales data. Today the data is moved by hand; we want to connect it **directly to MilkMaster** so it updates automatically — turning daily operations data into decisions, with zero manual work.

Executive summary

Where we are

A fully working platform running on real MilkMaster data — **487,000+ delivery records** across **Apr 2025** → **today**. Dashboards, AI insights and reports are live.

What we're asking

A **read-only** data connection from MilkMaster (an API key *or* an automatic daily file feed) so the platform stays current on its own.

Why it matters

Faster, data-driven decisions on products, hubs, pricing and retention — and a foundation for AI on top of the dairy's own numbers.

1 What we want

Today, someone exports a sales report from MilkMaster and uploads it into our platform by hand. It works, but it's manual and can fall behind. We want MilkMaster's sales/delivery data to flow into

Mr. Milk AI **automatically and continuously**. We do **not** need to change anything inside MilkMaster — only a read-only way to receive the data.

2 What our platform is

Mr. Milk AI OS is a private operating layer for the dairy business. It takes raw sales/delivery data and turns it into live dashboards, deep analytics, and an AI assistant that answers business questions — every answer backed by the real numbers.

Live analytics

Per-day product sales, hub-wise splits, monthly & seasonal trends, customers and revenue — all interactive.

AI assistant

Ask plain-English questions (“how did 1L milk sell in Wakad last month?”) and get answers computed from live data — no guessing.

Decisions & reports

Auto-generated, data-driven recommendations and one-click exports (Excel + an interactive dashboard report).

3 What we’ve built — and how

This is not a concept — it is running today on MilkMaster’s exported data.

487,000+

delivery line-items loaded

Apr 2025 →

now

date coverage

4,348

active customers

75

products

3

distribution hubs

₹4 Cr+

analysed (1L milk, 1 yr)

Capabilities already built

Automated import pipeline

Daily product sales

Detects overlap with existing data, removes duplicates, and appends safely — history is never wiped.

Any product / pack / date range, split by hub, with seasonality, weekday patterns and moving averages.

Hub-wise performance

Volume, revenue, share, customers and per-customer benchmarks for every distribution hub.

Reports & dashboard export

Multi-sheet Excel and a self-contained interactive dashboard with auto-written, data-driven recommendations.

Customer intelligence

Segmentation, retention/churn signals, campaign calendar and content tools.

Secure, multi-user, networked

Role-based access (owner / ops / marketing / CRM) and runs across the office network.

How it’s built (technical)

Layer	Technology	Purpose
Frontend	React + TypeScript (Vite), ECharts	Interactive dashboards & charts
Backend	FastAPI (Python)	Secure API & business logic
Analytics engine	DuckDB + pandas over columnar Parquet	Fast queries on 100k+ rows in <1s
Data store	PostgreSQL (Supabase) + Parquet	Customer master + transaction history
AI layer	Google Gemini agent (tool-use)	Answers grounded on a live data dictionary
Security	Role-based access, token sessions	Controlled, read-safe access

4 How this helps the business

The platform already surfaces concrete, money-relevant findings — computed only from the data, e.g.:

Weekday gap: Sundays sell ~10% below Mondays — closing the gap is worth an estimated **~5,500 L / year** in extra milk.

Hub headroom: Chinchwad averages ~100 L per customer vs Pune's ~116 — a clear upsell target.

Pricing: realised price is stable (~₹96/L, no erosion) — room to pilot a small premium tier.

Quality: disputed/spoilt is just **0.08%** of delivered — a strong reliability proof point for marketing.

With a live connection, these insights are always current — the dairy sees **yesterday's** performance every morning, not last month's.

5 What we need from MilkMaster

We don't need to know your database or internal technology. Any **one** of these read-only options lets us connect — please pick whichever is easiest:

1

An API + read-only key **BEST**

An endpoint we can call for sales/delivery history with a date filter. Fully automatic, always fresh, database-agnostic.

2

A scheduled file feed **EASIEST**

MilkMaster auto-exports a daily CSV/Excel to SFTP, Google Drive, S3 or email; our platform ingests it automatically.

3

Read-only database / replica access

Only if the above aren't possible — a read-only view of the relevant tables. (Here the database type matters.)

Questions for the technical engineer

1. Is there a **REST API**? If so: base URL, docs, and a **read-only key**.
2. Which endpoint returns **sales / delivery history**, and can we filter by **date range** (so we pull only new days)?
3. How is it **paginated**, and what are the **rate limits**?
4. If no API — can you **schedule an automatic export** (CSV/Excel) to a folder / Drive / S3 / email?
5. Could you share **one sample response** and confirm the fields below?
6. Is there a **sandbox/test** environment, and who is our **integration contact**?

Data fields we need

Delivery date

Customer (phone / id)

Product

Pack / weight

Quantity (delivered / net)

Amount (₹)

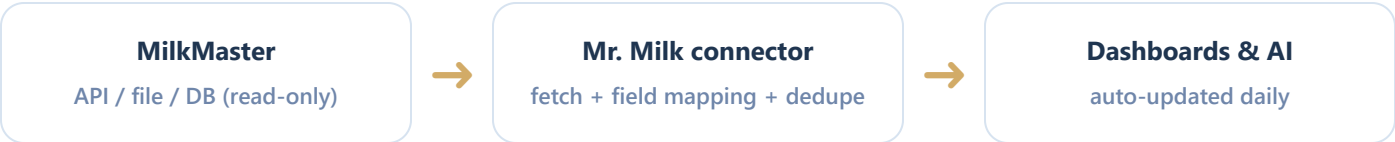
Delivery status

Hub / area

Invoice id

Nice-to-have: ordered/cancelled/returned quantities, delivery boy, route, subscription type, payment mode.

6 How the connection works



The connector pulls only **new** data since the last update, maps MilkMaster’s field names to ours, and feeds the existing import pipeline (which already handles de-duplication). The manual upload stays as a backup.

7 Security & ownership

Read-only. We only read data — never write to MilkMaster.

Private credentials. Keys live server-side, never in the app or shared.

Revocable. MilkMaster can disable access at any time.

Your data stays yours. It flows into Mr. Milk’s own systems; the AI runs on our backend.

8 Next steps & timeline

Step	Owner	Time
Choose a connection option & share details (key / feed / access)	MilkMaster	This week
Build the connector + field mapping	Mr. Milk	1–2 days
“Sync now” + nightly auto-sync	Mr. Milk	~1 day
Test on sandbox, verify totals match	Both	1 day
Go live — auto-updating dashboards	—	—

The ask for today: agree on **one** connection method (API key or automatic file feed preferred) and

identify MilkMaster's technical contact. Everything else is already built and ready.

Mr. Milk AI OS • Mittal Dairy Farms — Integration Brief for the MilkMaster meeting. All figures computed from live MilkMaster sales data.